Battery Energy Storage (BESS) Study — Peak Shaving, Backup, Markets, AI Smoothing

30 MW IT hyperscale campus, Dalarna (SE3), Sweden · fictional · conceptual study · 10 MW / 20 MWh at the reserved MSG-20 bays

Item	Value
Document	DWU-DC-E-CALC-004, Rev P1a (concept; errata on sheet 5) — Python package bessdc v0.1.0; LPs solved with HiGHS (scipy)
Basis	DWU-DC-E-DB-001 Rev P1, superseded by Rev P2 (40 MW subscription, MSG-20 BESS bay), CALC-002 (dcsizing PUE model, 37.8 / 42.6 MW), CALC-003 (relay settings, I''k)
Author / date	Donglin (Franklin) Wu — portfolio project 05 · 2026-10-25
Status	Concept — **not for construction or investment**. Public market data cited (accessed 2026-10-24), all other values [ASSUMED]; no employer/vendor data

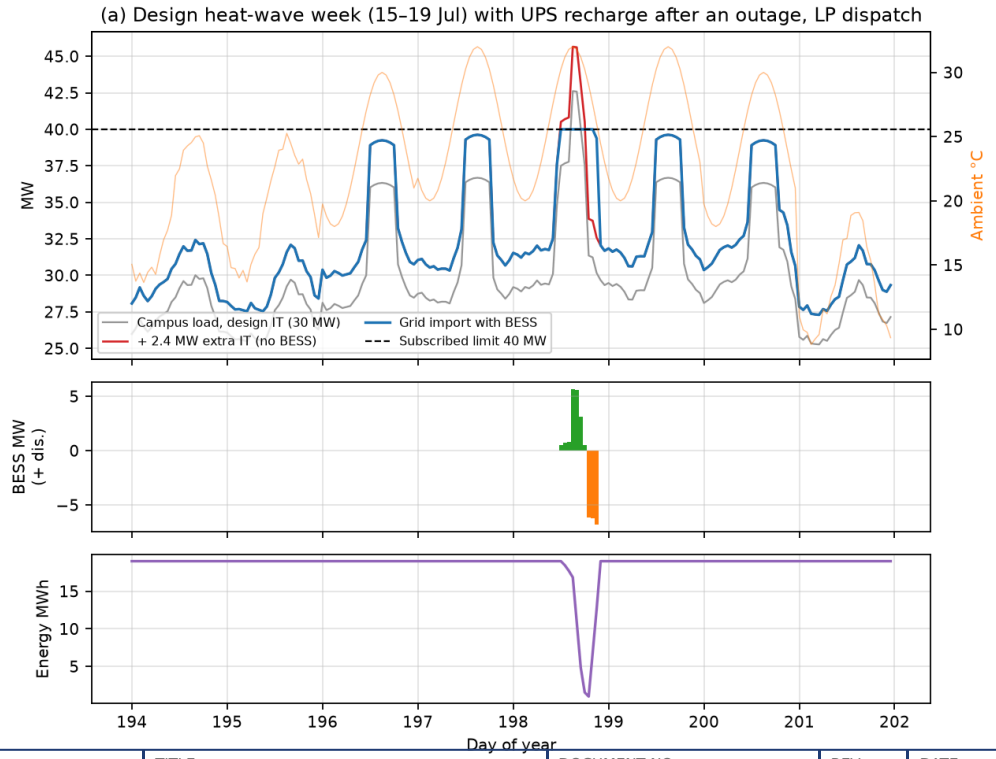
1. Scope and method

Hourly design year (8 760 h): synthetic Dalarna temperature (mean +4.5 °C, 5-day July heat wave to the +32 °C design ambient), IT utilisation (mean 0.80, diurnal/weekly, AI batch blocks, 100 % on heat-wave afternoons) × PUE(T) from the CALC-002 bin model, calibrated so 100 % IT at +32 °C = **37.8 MW**; UPS recharge after an outage (+4.8 MW, 1.5 h) placed on the hottest afternoon (CALC-002 campus max **42.6 MW**). Synthetic SE3 spot price scaled to the Nord Pool 2025 mean (46.23 €/MWh, ≈4 % hours ≤ 0); capacity prices = Svk Mimer annual means (2025, 2026 Jan–Aug, "saturated" = ½ × 2026 [ASSUMED]) with an hourly lognormal shape. **(a)** LP: max extra IT x s.t. $L_t + x \cdot s_t + c_t - d_t \le 40$ MW, SOC 5–95 %, η_{rt} 0.88, cyclic SOC. **(c)** LP co-optimising FCR-N, FCR-D up/down, mFRR CM up and behind-the-meter arbitrage with Svk LER endurance (FCR-N 1 h, FCR-D 20 min), power head-room, and the 40 MW limit including full down-activation. **(d)** 0.02 s simulation of a 12 MW synchronous AI training cluster. **(e)** 15-yr cash flow with fade/augmentation.

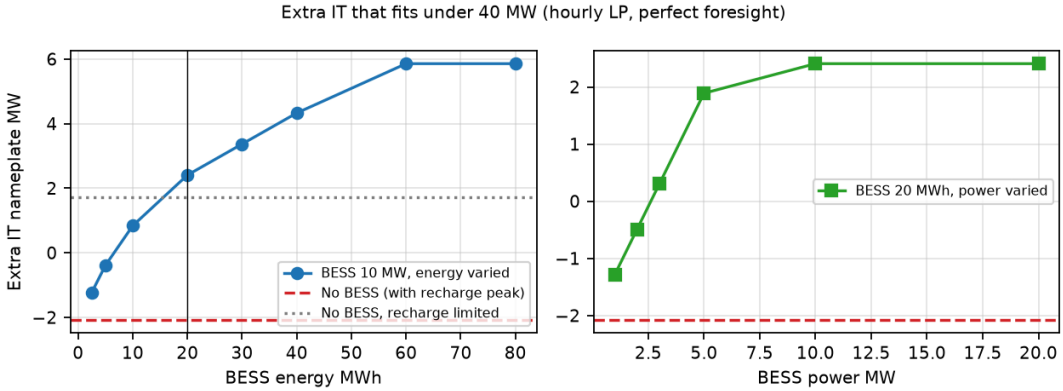
2. Key results

- **Grid limit is already exceeded without BESS** on the design day: 42.6 MW (incl. UPS recharge) vs 40 MW → headroom -2.08 MW IT. With recharge limited: +1.73 MW. The 10 MW / 20 MWh BESS gives **+2.41 MW IT** (4.48 MW gain) but only **+0.84 MW firm** with one 5 MW unit out. Peaks are few (7 h/yr) but long (heat-wave afternoons), so **energy, not power, binds**: 10 MW → 20 MW adds nothing; 20 → 40 MWh adds +1.9 MW.
- **Backup**: 16.9 MWh usable = 27 min at campus peak (and 10 MW < 37.8 MW) vs 24 h fuel. Replacing all 36 gensets would need ≈ 793 MWh (≈ 5.5 × genset CAPEX) and still not meet Tier III / EN 50600 Class 3. Island fault current 381 A at 20 kV: R-FDR would take ≈ 60 s → island protection needed.
- **Markets**: 1.82 M€/yr at 2026 prices (FCR-N 1.12, mFRR 0.45, arbitrage 0.19 M€), 4.15 M€ at 2025 prices, 0.96 M€ saturated. Honouring the 40 MW limit with max extra IT costs only **30 k€/yr** (1.6 %) → (a) and (c) are compatible if the EMS reserves head-room.
- **AI smoothing**: 3 MW/min ramp limit needs ≈ 11.4 MW but only 0.46 MWh; a 150 ms BESS leaves 5.6 MW p-p ripple in 0.1–2 Hz (limit 1.0 MW) — needs ≲ 10 ms effective response (20 ms → 1.26 MW). Hybrid with a 23 kWh / 10 MW supercap: 0.79 MW ✓.
- **Business case**: CAPEX 4.08 M€ → NPV 7.2 M€, IRR 34 %, payback 2.7 yr (2026 prices, −3 %/yr); saturated: NPV 2.0 M€, payback 5.3 yr. Capacity prices dominate; value is market risk, not engineering value. Each 100 k€/MW·yr of enabled-IT value adds 2.2 M€ NPV.

3. (a) Grid-capacity-limited peak shaving



PROJECT	TITLE	DOCUMENT NO.	REV	DATE	SHEET
D. Wu — Portfolio 30 MW IT DC campus, Dalarna SE3 (fictional)	**BESS study: peak shaving, backup, markets, AI** Concept — not for construction	**DWU-DC-E-CALC-004**	**P1a**	2026-10-25	1 of 5 (A4)



Extra IT (MW nameplate) vs BESS energy at 10 MW: 2.5 MWh → -1.23, 5.0 MWh → -0.39, 10.0 MWh → +0.84, 20.0 MWh → +2.41, 30.0 MWh → +3.37, 40.0 MWh → +4.33, 60.0 MWh → +5.86, 80.0 MWh → +5.86. Power at 20 MWh: 1 MW → -1.28, 2 MW → -0.49, 3 MW → +0.30, 5 MW → +1.89, 10 MW → +2.41, 20 MW → +2.41. Perfect foresight: real EMS needs a peak forecast and must hold SOC high before hot afternoons; firm capacity assumes the N-1 BESS (one 5 MW / 10 MWh unit out of service). Alternative levers without BESS: staggered UPS recharge (≤ 2.2 MW) and IT power capping in the few design hours.

4. (b) Backup role — what a BESS can and cannot credibly replace

Function	Credible?	Reason (concept level)
Replace all 36 gensets (Tier III / EN 50600 Class 3)	**No**	Tier III needs on-site engine generation able to carry the design load without runtime limit; a battery with finite energy is not accepted as the generation source. 24 h autonomy needs ≈ 0.85 GWh of storage.
Replace the static UPS (ride-through of the grid→genset gap)	**No**	BESS sits at 20 kV; the gap occurs at LV behind the ATS (open transition) and inside a ~10 ms window. UPS also gives per-block redundancy (2N/DR) that one MV BESS cannot.
Reduce number/rating of gensets (N+1 → N)	**Not recommended**	DR topology has no stand-alone redundant genset; derating gensets on the assumption that the BESS covers peaks fails when the BESS is out of service or depleted (single point of dependency).
Cover a genset that fails to start (bridging, grid-forming island)	Possible (informational)	Grid-forming PCS energises the 20 kV island; blocks whose ATS stays on the transformer side are fed for hours. Needs islanding scheme, soft energisation, island protection (51V/27/87L) and earthing in island.
Genset load-step / start support, recharge-peak cover	Yes	Only when the BESS and gensets share a bus (not in current LV-genset topology) or via the 20 kV island; grid-connected recharge peaks (42.6 MW > 40 MW) are covered — see (a).
Genset testing on load (replace load banks), HVO fuel saving	Yes	Monthly test runs can be loaded by charging the BESS (if synchronised); fewer dummy-load hours.
Black start / re-energisation of MSG-20 after grid return	Possible	Soft energisation of 20 kV transformers limits inrush; closed-transition re-transfer reduces UPS cycling.

Tier III / EN 50600. Uptime Institute Tier III treats on-site engine generation as the primary power source that must carry the design load with no runtime limit; EN 50600-2-2 Class 3/4 needs an independent secondary supply. A finite-energy battery does not satisfy this, so gensets stay N-rated per DR block. **Ratings** (ISO 8528-1): ESP (standby, limited hours) is not enough alone; PRP/COP and data-centre ratings with unlimited hours at variable load are accepted — DB-001 uses 2 750 kVA / 2 200 kWe DCC (manufacturer's data-centre continuous rating). **Start-up**: gensets reach load in ≈ 10–15 s (UPS bridges this); the BESS at 20 kV cannot reach the LV bus behind an open-transition ATS.

Island fault clearing with the CALC-003 settings (BESS 381 A at 20 kV = 1.2 × I_n of 11 MVA)

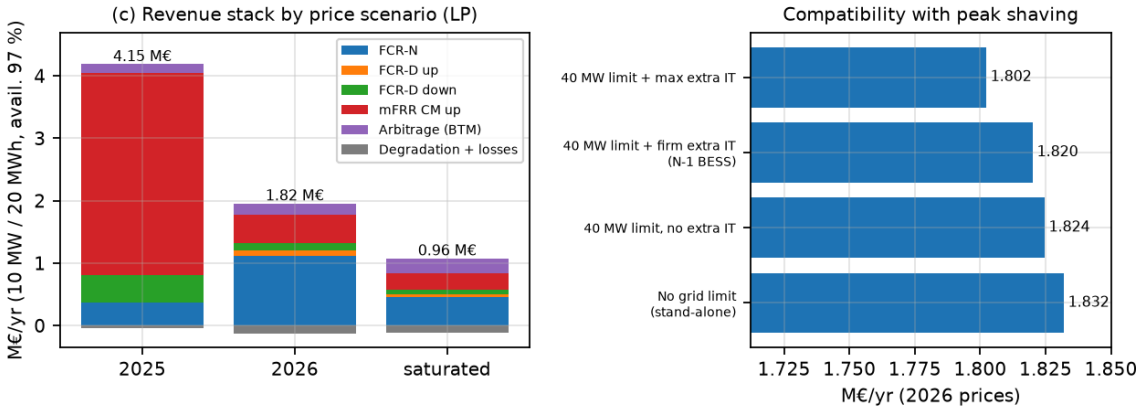
Device	I fault (A)	Pick-up (A)	× pick-up	t clear (s)	≤ 1 s
R-FDR hall feeder 20 kV	381	320	1.19	59.50	✗
R-INC 20 kV incomer	381	2 100	0.18	∞	✗
R-TX block transformer 20 kV	381	100	3.81	5.98	✗
MSB incomer ACB (LV)	18 145	3 600	5.04	0.35	✓
Busway feeder MCCB (LV)	10 887	800	13.61	0.02	✓

Grid-connected the BESS adds +0.38 kA: MSG-20 13.0 → 13.3 kA (worst abnormal 22.7 → 23.1 kA vs 25 kA, 92 %). In an inverter-fed island the 20 kV inverse-time relays are too slow or never pick up; LV devices still trip (fault current referred to LV ≈ 18.1 kA). Island mode would need 51V/27 definite-time, 87L or a fault-current-boosting grid-forming PCS. LV-side ACBs clear in 0.35 s but the whole island voltage collapses during the fault (UPS on battery).

Fire safety (concept)

NFPA 855 (2026 ed.): Installation of ESS: size/separation limits, explosion control, detection, emergency response plan; outdoor separation ≈ 3 m (10 ft) unless reduced by large-scale fire test data · **UL 9540 / UL 9540A**: System safety listing / cell-module-unit-installation thermal-runaway propagation test · **IEC 62933-5-2**: Safety requirements for grid-integrated electrochemical EES systems · **EN IEC 62485-5**: Safety for stationary Li-ion batteries (installation, ventilation, separation) · **IEC 62619 / IEC 63056**: Cell and battery safety for industrial Li-ion applications · **NFPA 68 / NFPA 69**: Deflagration venting / explosion prevention (gas detection + exhaust) · **Swedish context**: LSO (Lag om skydd mot olyckor), BBR (Boverket) fire separation, MSB guidance on battery storage, rescue-service consultation; insurer rules (e.g. FM DS 5-33).
Design intent: Outdoor containerised LFP units, ≥ 10 m from data halls, 130/20 kV transformers and fuel tanks [ASSUMED project choice]; Per-container gas (H_2/CO) detection, deflagration vents or NFPA 69 exhaust, no walk-in during thermal event; Water supply for defensive cooling of neighbours, no in-container extinguishing reliance; Fire-rated cable routes and separate 20 kV bays/sections so a BESS fire cannot take out both MSG-20 sections; UL 9540A data for the selected product to justify separation; emergency response plan with Räddningstjänst.

5. (c) Revenue stacking — FCR-N / FCR-D / mFRR / arbitrage



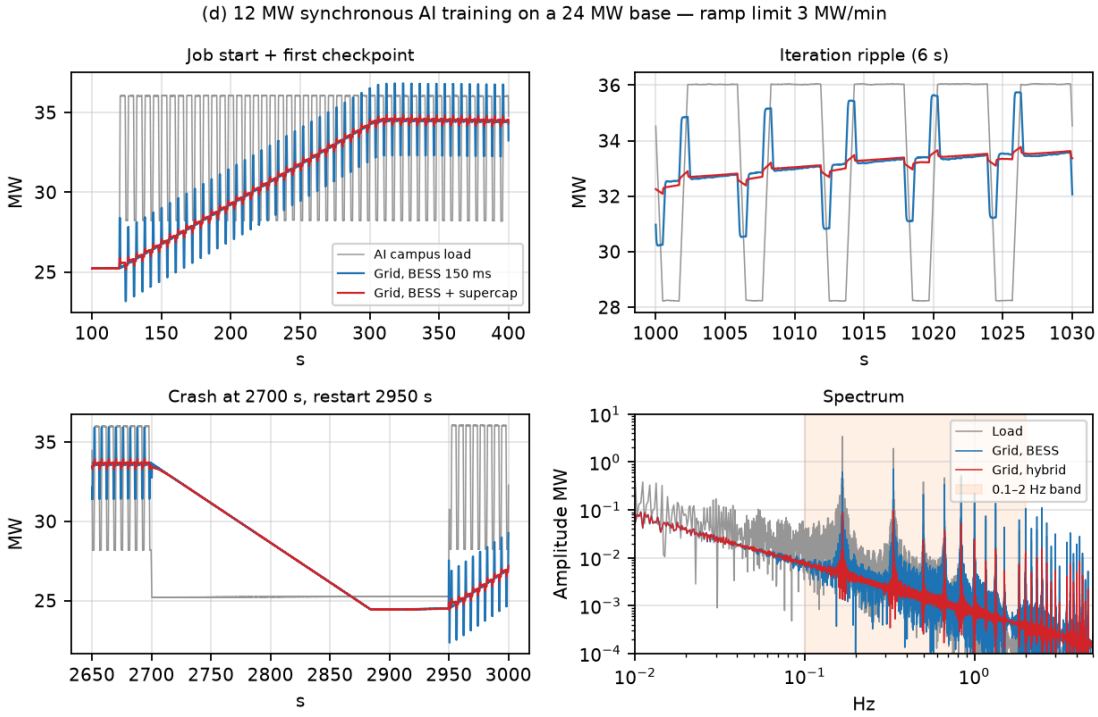
k€/yr (avail. 97 %)	2025	2026	saturated
FCR-N	369	1 122	463
FCR-D up	4	76	42
FCR-D down	432	121	70
mFRR CM up	3 241	451	258
Arbitrage (BTM)	140	186	243
Degradation (15 €/MWh)	-20	-31	-41
FCR-N activation losses	-20	-101	-76
Total	**4 146**	**1 824**	**959**
Equivalent full cycles /yr	124	352	338

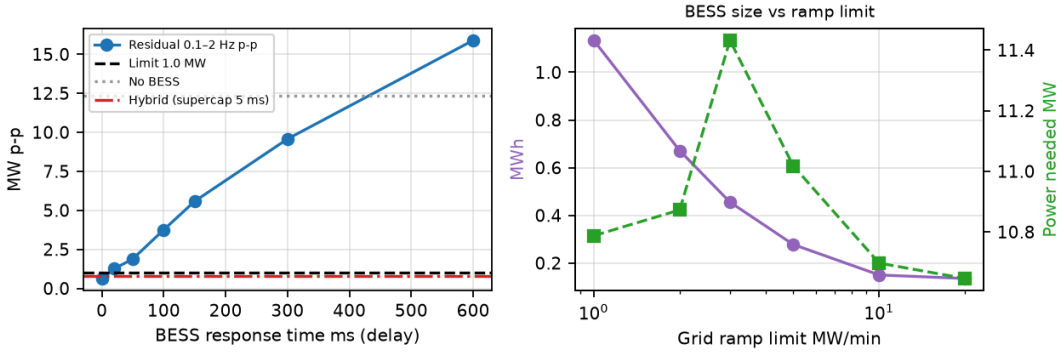
€/MW/h (mean)	2025	2026	sat.
FCR-N	26.8	22.1	11.1
FCR-D up	6.1	5.6	2.8
FCR-D down	5.9	3.2	1.6
mFRR CM up	40.9	11.3	5.7

Svk Mimer annual means via Flexnavet (secondary); Svk monthly FCR report Dec 2025 (FCR-N 16.3, FCR-D up 5.0 €/MW/h) and the Svk Balancing Market Outlook 2030 confirm falling prices as BESS volume grows.

Perfect-foresight LP is an upper bound → ×0.85 realisation in (e). FCR-N throughput ≈ 10 % activation [ASSUMED]; down-reserve activation imports energy and is counted against the 40 MW limit (the compatibility constraint with (a)); no export assumed (behind the meter). Market depth not modelled: FCR volumes are a few hundred MW nationally, so revenue per MW does not scale to large fleets. mFRR CM up 2025 (40.9 €/MW/h) collapsed in 2026 (11.3) — this alone moves the 2025 total by ≈ 2.8 M€.

6. (d) AI-training power-swing smoothing

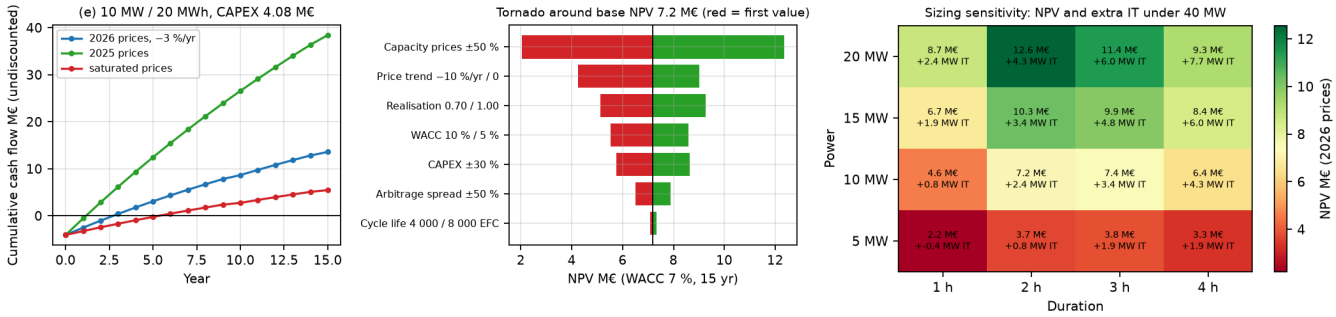




Base 24 MW + 12 MW AI: 6 s iterations (compute 1.0 / all-reduce 0.35 pu), 0.5 s facility rise time, checkpoints every 15 min (45 s at 0.2 pu), job start, crash and restart [ASSUMED, after NERC Large Loads observations]. Controller: LPF (30 s) + SOC feedback + 3 MW/min rate limiter; actuator = transport delay + 20 ms current loop. Residual ripple vs response: 0 ms → 0.63, 20 ms → 1.26, 50 ms → 1.89, 100 ms → 3.75, 150 ms → 5.58, 300 ms → 9.57, 600 ms → 15.89 MW p-p. Slow response (≥ 300 ms) is **worse than no smoothing** for the ripple (delayed injection). Smoothing throughput ≈ 1.6 MWh/h ≈ +531 EFC/yr at 6 000 h training → shortens life.

Option	Response	Energy	Cycle life	Comment
Li-ion BESS at 20 kV (this study)	100–300 ms	Minutes–hours	≈ 6 000 EFC (shallow µ-cycles less severe)	Shares asset with peak shaving/FCR; must reserve P/E for smoothing; fast oscillations add throughput
Supercapacitor / hybrid (EDLC)	< 10 ms	Seconds (≈ 1–30 s)	≈ 10^6 cycles	Ideal for the 0.1–2 Hz iteration ripple; cannot cover crash/start ramps (needs minutes)
UPS battery (LV, per block)	≈ ms (already online, double conversion)	≈ 5 min at full IT load	Li-ion UPS ≈ 3 000–6 000 cycles	Closest to the load; but cycling erodes backup reserve and warranty; requires UPS firmware support (grid-interactive mode)
GPU / software power floor ('burn' kernels, power capping)	Instant	n/a (wastes energy)	n/a	Cheapest; costs energy (e.g. 5–15 % of AI power) and needs workload-owner cooperation

7. (e) Business case and sizing sensitivity



CAPEX = (120 €/kWh·E + 75 €/kW·P) × 1.20 BoP + 0.3 M€ bays [ASSUMED; ≈ 157 €/kWh turnkey at 2 h vs BNEF 2025 Europe 177 USD/kWh]. OPEX 2 %/yr, WACC 7 %, 15 yr, ancillary prices −3 %/yr, fade 1 %/yr + cycles (6 000 EFC to 80 %), augmentation to 90 % when SOH < 80 %, avoided subscription 2.6 MW × 242 SEK/kW·yr = 58 k€/yr. NPV scales ~linearly with power because FCR is paid per MW — but only while the market is not saturated; 2 h is the sweet spot (FCR-N needs 2 MWh/MW, more energy mainly buys extra IT).

8. Conclusions and design changes (→ TODO_final_fixes.md B22–B31)

#	Item	Phase 1–3	Project 05
B22	20 kV BESS bays	bay reserved	2 × 630 A CB bays (5 MW each); 50/51, 67N, 27/59, 81, ROCOF, intertrip, 25
B23	SLD-001	bay symbol	Draw CB/CT/VT/SA, PCS transformers Dyn 2 × 5.5 MVA, metering
B24/25	MSG-20 I''k / Ic	13.0 kA / 22 A	13.3 kA (worst 23.1 kA, 92 %); Ic +≈1 A
B26	UPS recharge vs 40 MW	assumed inside	42.6 MW > 40 MW → limit/stagger recharge or BESS cover
B27	EMS priorities	—	Import limit → SOC reserve → down-reserve head-room → markets; firm IT only N-1
B28	Backup concept	—	BESS not counted for Tier III; island mode needs 51V/27/87L + grid-forming
B29–31	Q/harmonics, fire, AI	—	Harmonic study; ≥ 10 m separation, UL 9540A; ramp clause + fast smoothing

Limitations. Synthetic profiles and prices (one weather/price year, perfect foresight); hourly LP misses intra-hour FCR activation and SOC drift; no market depth, bidding or aggregator model; no export or DSO tariff for BESS; generic PCS fault current (1.2 pu) and response; AI profile is illustrative, not measured; grid-code ramp/oscillation limits are [ASSUMED] (no Svk/DSO limit exists yet for this case); Tier III/EN 50600 statements are interpretations at concept level, not certification advice. Code: `scripts/make_all.py` regenerates figures/results; 23 pytest tests incl. Streamlit AppTest.

Revision history and errata

Rev	Date	Description
P1	2026-10-24	First issue
P1a	2026-10-25	Errata only: cross-references to DB-001 Rev P2 / SLD Rev P2; superseded statements listed below. No calculation result changed.
P1a, ed.	2026-10-29	Editorial: genset rating wording on sheet 2 aligned to DB-001 (2 750 kVA / 2 200 kWe DCC, previously "DCP"). No calculation result changed.

Note (Rev P1a, 2026-10-25). Results are unchanged. Changes B22–B31 are implemented in **DB-001 Rev P2** §7.5–7.6 and the new drawing **SLD-004** (BESS plant), with bays Q19 / Q29 on SLD-001 sh 2. The I''_k values here supersede the "no BESS contribution" limitation in CALC-003.